

Probability Calculation for Natural frequency of Anisotropic Stator Core of Large Turbo-Generator

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Abstract: - Techniques for vibration reduction and design of stator core require knowledge of its modal frequencies, which depend on the geometry shapes, dimensions and material properties. The material properties of stator core are mainly related to its laminated process and assembly as well as the operating temperature of turbo-generator. It is found that the generally accepted value of material properties is not valid for the stator core with laminated silicon steel slice and operating at about 110-120 Celsius. This paper simply summarizes the method of determining elastic constants of stator core material, and then introduces a simple and effective method for the calculation of the modal frequencies of stator core. The material model of stator core is thought anisotropic. The elastic modulus, Poisson's ratio, and mass density are all considered as uncertainty variables, obeying Gaussain distribution, and changing in the possible range; and then ANSYS workbench 14.5 program is used to determine the resonant frequencies and corresponding probability of stator core ,and response surfaces of natural frequencies vs. elastic constants by statistical method (Six Sigma Analysis). The results have an important guiding role for the laminated structure design, manufacture process, the fixedness and the vibration isolation design of stator core.

Key-Words: - Large turbo-Generator, Anisotropic Stator Core, Material Properties, Natural Frequency, Six Sigma Analysis

1 Introduction

1.1 Previous Works

Vibration and acoustic noise in the large turbo-generator is mainly caused by the oval deformation of the stator lamination stack due to its radial magnetic attraction to the rotor [5, 6, 9]. There are several ways of determining stator core resonant frequencies and mode shapes of the large turbo-generator: analytical calculations[7,8,10,18], numerical computation[1,2,11,12] and experimental measurements[17].

The paper [15] proposes a measurement method of the Young modulus, shear modulus, mass density for stator core of switched reluctance motor. However, the Young modulus value of stainless steel ($E=2.07 \times 10^{11} \text{N/m}^2$) has been widely used by researchers in small electric machine vibrations area.

The fact is that the stator lamination steel of a large turbo-generator is different from stainless steel; in addition, the material properties may change during core lamination, manufacturing, and vary with operation temperature of turbo-generator. In fact, the Young modulus value of stator core varies with exciting current, temperature and structure shape. In other words, the elastic constants and mass density of the stator core is actually uncertain

variables, distributed in an interval, subject to a statistical distribution law; so the natural frequency value for the linear system should be subject to the same distribution. If Finite Element Method (FEM) is used to model, the elements of mass matrix and stiffness matrix are random variables, and the conventional method cannot be used for the calculation of natural frequency of stator core, but the method of probability statistics; for stator core of large turbine generator, the literature in this area are rare. Because of the complexity of the material characteristics of stator core, in the calculation of the natural frequency of the stator core of large turbine generator, the all researchers consider that the material constants are unchanged. This paper is an in-depth research.

In the author's paper[3, 4], we consider that the stator core for large turbo-generator is isotropic body ,so the calculation of mechanical characteristic needs only the elastic modulus and Poisson's ratio, and material properties is considered as uncertain variables .

In fact, the stator core assembly of large turbo generator is an anisotropic body. A new nonlinear anisotropic model for soft magnetic materials is proposed for determining electro-magnetic properties of electromagnetic device [13]and

$$G_{13} = \frac{E_1}{2(1+\nu_{13})} = 0.65 \times 10^5 \text{ N/mm}^2$$

Number of stator slots is 36; Outer diameter of stator core 2945 mm; Inner diameter of stator core 1471mm; Yoke inner diameter of stator core 1791mm; Length of stator core 8100 mm; Height of slot 160 mm;.

Parameters of equivalent cylindrical shell of the stator core of 1000MW turbo-generator are as follows: Radius of middle surface is 1184 mm; Thickness 577 mm; Length of shell 8100 mm; Material constants are above.

In the dynamics calculation of stator core, the shell181 and solid185 element are selected, isotropic and orthotropic material model is adopted respectively. In the ANSYS/Workbench software, the certainty dynamics equation is as follows:

$$([K] - \omega^2 [M])\{u\} = 0 \quad (16)$$

Where:

$[M]$, $[K]$ is mass matrix and stiffness matrix respectively. $\omega, \{u\}$ is natural frequency and mode shape respectively. Obviously, these relate with geometric dimensions and material parameters. In our problem, the geometric dimensions are constant, and material parameters are variation.

The results of natural frequencies are shown in Table 1, and the mode shapes for isotropic stator core can also see [1-3]. The results show that thick shell model and the 3D solid model can be used as the calculation of the stator core model. The results of the two models differ little, comparatively speaking, a thick shell model is simpler, and fewer elements, nodes and the number of degrees of freedom, less computer resources and consumption time required, faster speed of calculation. This model is ideal for scanning to optimize the design; and 3D solid model only as a reference. The modal experiments of stator core prototype is quite difficult, so far we have not seen the relevant literature about these of 1000MW generators. We just use a different model (the number of nodes and elements) for comparison, and compare with the calculation results of different software platforms, so that the accuracy of calculations is ensured. The FEM solution of thick shell model have high precision as comparison with the results [17, 18], and is available for stator core.

Table 1. Natural Frequencies for Stator Core (Hz)

Freq. Oder	FEM solution			
	Thick shell		3D solid	
	Isotropic	Orthotropic	Isotropic	Orthotropic
1-6	0	0	0	0
7	142.16	150.91	144.5	155.83
8	142.16	150.92	144.5	155.83
9	168.85	179.41	167.34	182.54
10	201.4	221.88	206.68	229.12
11	201.5	221.88	210.61	233.36
12	204.77	226.47	211.51	235.31
13	204.77	226.47	215.06	248.25
14	230.69	255.62	240.41	264.27
15	263.45	255.62	266.6	270.42
16	263.45	279.98	272.3	280.23

4 Dynamics statistical model and calculation of stator core

4.1 Six Sigma Analysis Method [19]

In traditional deterministic analyses, uncertainties are either ignored or accounted for by applying conservative assumptions. Uncertainties would be typically ignored if the input parameter has no effect on the behavior of the component under investigation. In this case, only the mean values or some nominal values are used in the analysis. However, in some situations, the influences of uncertainties exist, and cannot be neglected, as for the material properties of stator core, for which the scatter is not usually ignored.

A Six Sigma Analysis uses statistical distribution functions (such as the Gaussian or normal distribution, the uniform distribution, etc.) to describe uncertain parameters, and allows us to determine the extent to which uncertainties in the model affect the results of an analysis.

In a deterministic analysis, computer models are described with specific numerical and deterministic values. The accuracy of a deterministic analysis depends upon the assumptions and input values used for the analysis.

While scatter and uncertainty naturally occur in every aspect of dynamics analysis of stator core, deterministic analyses do not take them into account. To deal with uncertainties and scatter, use Six Sigma Analysis, which we can use to answer the following questions: when the input variables of a finite element model are subject to scatter, how large is the scatter of the output parameters? How robust are the output parameters? When the output is subject to scatter due to the variation of the input variables, then what is the probability that a design criterion given for the output parameters is no longer met? How large is the probability that an unexpected and unwanted event takes place (i.e., what is the failure probability)? Which input